

Dynamic Cooper and John Economies: The Role of Coordination Failures and Accumulation in Equilibrium Emergent Phenomena

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Abstract

We study the type of equilibria that emerges when one increases the level of interactions between agents in a dynamic extension of the Cooper and John's [1988] model. In the model, agents exert an effort that accumulates into a stock. The payoff function V depends on individual effort, individual stock, but also possibly on the aggregate level of effort, as capturing interactions between agents. Three features of the payoff function V will be relevant: how does the aggregate level of effort impact marginal value of individual effort or marginal value of individual stock; whether interactions are under the form of complementarities or substitutabilities; and finally whether past accumulation increases or decreases current incentives to make effort. Depending on those properties, we show how hysteresis, limit cycles and sunspots can emerge from interactions.

Key Words: Emergence, Coordination, Hysteresis, Limit Cycles, Sunspots;
JEL Class.: E3

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Introduction

In the seminal work of Cooper and John [1988], the authors showed how coordination failures can motivate “Keynesian” macroeconomics. In particular, in their simple (but nonetheless quite general) static framework, strategic complementarities between agents (i) causes equilibrium economic activity to be inefficiently low and multiplier effects to emerge,¹ and (ii) opens the door to the possibility of multiple equilibria, which cannot occur in their absence.

While clearly important, the properties considered in Cooper and John [1988] are explicitly static in nature. In the macroeconomic literature, however, much emphasis has been put on how strategic interactions can affect the *dynamic* properties of macro models. For example, numerous business cycle models have been proposed in which coordination failures create “emergent” phenomena, i.e., dynamic equilibrium properties that are fundamentally driven by the effects of strategic interactions. Examples of such emergence include indeterminacy (e.g., as in Benhabib and Farmer [1994]), hysteresis (e.g., as in Taschereau-Dumouchel and Schaal [2015]), and limit cycles (e.g., as in Beaudry, Galizia, and Portier [2020]).

In this paper, we extend the abstract setup of Cooper and John [1988] to a dynamic environment, and use it to try to understand the fundamental properties of agents’ payoff functions that are crucial to producing different forms of emergent phenomena.

The paper is organized as followed. We first study cases in which restriction on parameters imply that the dynamics is of order one. We then fully characterise the set of bifurcations when $V_{23} = 0$ or $V_{13} = 0$. Finally, we provide an encompassing DSGE type model in which we numerically investigate the set of emergent equilibria.

1 Model and Method of Analysis

1.1 Model

We consider a deterministic world in which there is a large number of identical agents indexed by i . In period t , each agent chooses private action e_{it} , given the average action of others

¹For simplicity, we assume throughout that the type of strategic interaction (none, complementarity, or substitutability) is the same globally in the strategy space (e.g., if there is strategic complementarity at some point in the strategy space, then there is strategic complementarity everywhere). While potentially interesting, allowing for different regions characterized by different types of interaction would considerably complicate the discussion.

e_t and the value of a private predetermined stock variable X_{it} that aggregates past actions according to

$$X_{it+1} = (1 - \delta)X_{it} + e_{it}, \quad (1)$$

with $0 \leq \delta \leq 1$. The payoff function of agent i in period t is then given by $V(e_{it}, X_{it}, e_t)$, which is assumed to be concave in (e_{it}, X_{it}) and three times continuously differentiable in all arguments. To use Cooper and John's [1988] terminology, there will be spillovers when $V_3 \neq 0$ and strategic interactions when $V_{13} \neq 0$ and/or $V_{23} \neq 0$. For example, we may embed a Walrasian economy—in which prices, which are a function of aggregate effort e , mediate strategic interactions—in this environment by taking, e.g., $V_{23} > 0$ and $V_{13} < 0$: additional demand for goods by others raises prices, which, all else equal, raises the value of an individual's own existing stock of goods, while simultaneously causing that individual to demand fewer new goods themselves. In addition to price-mediated ones, spillovers and strategic interactions could also come from, for example, thick-market externalities (as in search models), direct productive externalities (as in endogenous growth models *à la* Aghion and Howitt [1992]), or aggregate demand externalities (as in many models featuring monopolistic competition).

Agent i is forward looking and, given the sequence of e_t and X_t , chooses the sequences of e_{it} and X_{it+1} that maximizes

$$\sum_{t=0}^{\infty} \beta^t V(e_{it}, X_{it}, e_t),$$

with $0 \leq \beta < 1$, given X_{i0} and subject to (1) and a transversality condition

$$\lim_{T \rightarrow \infty} \beta^T V_2(e_{iT}, X_{iT}, e_T) X_{iT} = 0. \quad (2)$$

The Euler equation of that problem writes

$$V_1(e_{it}, X_{it}, e_t) = \beta \left((1 - \delta) V_1(e_{it+1}, X_{it+1}, e_{t+1}) - V_2(e_{it+1}, X_{it+1}, e_{t+1}) \right). \quad (3)$$

We will focus on symmetric allocations, and will therefore study the properties of equilibrium

allocations given by

$$V_1(e_t, X_t, e_t) = \beta \left((1 - \delta) V_1(e_{t+1}, X_{t+1}, e_{t+1}) - V_2(e_{t+1}, X_{t+1}, e_{t+1}) \right), \quad (4)$$

$$X_{t+1} = (1 - \delta)X_t + e_t, \quad (5)$$

$$\lim_{T \rightarrow \infty} \beta^T V_2(e_T, X_T, e_T) X_t = 0, \quad (6)$$

for X_0 given.

1.2 Method of Analysis

Our goal is to study how does the dynamics of equilibrium allocations be affected by the presence of interactions between agents. We start from a situation in which, absent of interactions (which corresponds to a world with isolated agents), the equilibrium is determinate and smoothly converges to its unique steady state (denoted (e^s, X^s)) in a non-oscillating way. We are then interested in the aggregate dynamics once agents interact; in which case the dynamics of the whole might qualitatively differ from the dynamics of each of its part as taken separately.

For that purpose, we make use of the theory of bifurcations. In dynamical systems, a bifurcation occurs when a small smooth change made to the parameter values (the bifurcation parameters, V_{13} or V_{23} in our setup) of a system causes a sudden “qualitative” or topological change in its behaviour. Generally, at a bifurcation, the local stability properties of equilibria, periodic orbits or other invariant sets changes.

The local stability properties is analyzed by considering the linear approximation of the dynamics around the no-interactions steady state, that is given by

$$\begin{aligned} & \left(V_{11}(e^s, X^s, e^s) + V_{13}(e^s, X^s, e^s) \right) e_t + V_{12}(e^s, X^s, e^s) X_t \\ & = \beta \left((1 - \delta)(V_{11}(e^s, X^s, e^s) + V_{13}(e^s, X^s, e^s)) - (V_{21}(e^s, X^s, e^s) \right. \\ & \quad \left. + V_{23}(e^s, X^s, e^s)) \right) e_{t+1} - \beta \left((1 - \delta)V_{12}(e^s, X^s, e^s) - V_{22}(e^s, X^s, e^s) \right) X_{t+1}, \end{aligned} \quad (7)$$

and by accumulation equation (1), with $\lim_{T \rightarrow \infty} \beta^T V_2(e_T, X_T, e_T) X_t = 0$ and X_0 given. Note that when there is no ambiguity, we write V_{ij} instead of $V_{ij}(e^s, X^s, e^s)$.

Varying V_{13} or V_{23} , we theoretically derive the behaviour of the model's eigenvalues, and show that some might enter or exit the unit disk (hence a bifurcation), in which case hysteresis, sunspots equilibria or limit cycle will emerge.

1.3 Assumptions

We make a set of “regularity assumptions” to guarantee that the no-interactions model is determinate with non-oscillating transition to a unique steady state. Therefore, indeterminacy, oscillations or instability will be only the possible consequence of introducing interactions. The first assumption we make is equivalent to assuming that households have a smoothing motivation for e .

Assumption A1. $\chi \equiv (1 - \delta) V_{11} - V_{12} < 0$.

Assumption A1 ensures that, given X_{it} , and holding X_{it+1} and the aggregate variables constant, an increase in e_{it+1} would cause the household to want to increase e_{it} .²

Note that this assumption does not directly impose that an increase in e_{it+1} will cause the household to choose a higher e_{it} (holding the aggregate variables constant), since a higher e_{it} would cause X_{it+1} to increase, while Assumption A1 was derived holding X_{it+1} constant. However, in combination with the other restrictions made (i.e., $V_{11}, V_{22} < 0$ and $\beta, \delta \in (0, 1)$), Assumption A1 will indeed ensure this. In particular, for future reference, note that Assumption A1 implies

$$\theta \equiv V_{11} - \beta(1 - \delta)V_{12} + \beta V_{22} < 0. \quad (8)$$

The second assumption is related to the the no-interactions steady state. Given equations (4) and (5), X is a steady state if satisfies $S(X) = 0$, where

$$S(X) = [1 - \beta(1 - \delta)] V_1(\delta X, X, \delta X) + \beta V_2(\delta X, X, \delta X).$$

Note that

$$S'(X) = \underbrace{[1 - \beta(1 - \delta)] \delta V_{11} + [1 - \beta(1 - 2\delta)] V_{12} + \beta V_{22}}_{\kappa(X)} + \underbrace{[1 - \beta(1 - \delta)] \delta V_{13} + \beta \delta V_{23}}_{\psi(X)}.$$

²In particular, totally differentiating (3) with respect to e_{t+1} (holding X_{it} , X_{it+1} , e_t , and e_{t+1} constant), we obtain that $de_{it}/de_{it+1} > 0$ if and only if A1 holds.

In the above expression, $\kappa(X)$ (or κ for short) is the the slope of the S function when there are no interactions, in which case $\psi(X) = 0$ (or ψ for short), while $\psi(X)$ is the contribution of interactions to the slope of S .

Assumption A2.

(i) $S(X) = 0$ has at least one solution when there are no interactions – ie when $V_{13} = V_{23} = 0$,³

(ii) $\kappa = [1 - \beta(1 - \delta)]\delta V_{11} + [1 - \beta(1 - 2\delta)]V_{12} + \beta V_{22} < 0$.

Under Assumption (A2) (ii), the no-interactions steady state is unique. We denote (e^s, X^s) this unique no-interactions steady state. To ease the exposition, we make the extra assumption that, without loss of generality, (e^s, X^s) is also a steady state when there are interactions – ie when $\psi(X^s) \neq 0$. In words, interactions will only affect the *slope* of the S function at the no-interactions steady state. Note also for future reference that Assumptions A1 and A2 together imply that

$$\frac{\chi}{\theta} = 1 - \frac{\kappa + \beta\delta\chi}{\theta} < 1. \quad (9)$$

It is also the case that under Assumption A2 the steady state exists and is unique when there are substitutabilities ($\psi(X) < 0$), as $S'(X) = \kappa + \psi < 0$. When $\psi(X) > 0$ – ie when we have complementarities, there could be a range of X such that S slopes positively when complementarities are strong enough at X^s , more precisely when $\psi(X^s) > |\kappa(X^s)|$. Under assumption A3 that states that interactions die out at zero and infinity, then there will be necessity (at least) two new steady states that will appear, one (X_L) below the no-interactions steady state and one (X_H) above.

Assumption A3. *Interactions die out at zero and infinity:*

(i) $\lim_{e \rightarrow 0} V_{13}(\cdot, \cdot, e) = 0$, $\lim_{e \rightarrow 0} V_{23}(\cdot, \cdot, e) = 0$,

(ii) $\lim_{e \rightarrow +\infty} V_{13}(\cdot, \cdot, e) = 0$, $\lim_{e \rightarrow +\infty} V_{23}(\cdot, \cdot, e) = 0$.

³One could derive sufficient conditions for the existence of at least one steady state, but such conditions are not particularly interesting.

Next, to a first-order approximation around the steady state, equation (3) can be written as is given by

$$\widehat{e}_{it} = \alpha_0 \widehat{e}_{it+1} + \alpha_1 \widehat{X}_{it} + \gamma_0 \widehat{e}_t + \gamma_1 \widehat{e}_{t+1},$$

where

$$\begin{aligned} \alpha_0 &\equiv \frac{\beta \chi}{\theta} \quad , \quad \alpha_1 \equiv -\frac{\sigma}{\theta} \quad , \\ \gamma_0 &\equiv -\frac{V_{13}}{\theta} \quad , \quad \gamma_1 \equiv \frac{\beta [(1-\delta) V_{13} - V_{23}]}{\theta}, \end{aligned}$$

with

$$\sigma \equiv [1 - \beta (1 - \delta)^2] V_{12} + \beta (1 - \delta) V_{22}.$$

Given that $\theta < 0$, the denominators of all these expressions are negative. From Assumption A1 we thus see that $\alpha_0 > 0$. Further, as we would expect, $\gamma_0 > 0$ if and only if $V_{13} > 0$. We also see that the sign of α_1 –i.e., of the direct effect of $\widehat{X}_{it}$ on $\widehat{e}_{it}$ – is the same as the sign of σ . Note that it is possible to get $\sigma < 0$ while still having $V_{12} > 0$; that is, we don't need substitutability between e_{it} and X_{it} in V in order to have $\alpha_1 < 0$ as long as V is sufficiently concave in X .

We make Assumption A4 so that period- t equilibrium (for given expectations) is unique. Indeed, we want to rule out this source of static multiplicity in the model (as for example modelled in Fernandez-Villaverde, Mandelman, Yu, and Zanetti [2019]), and focus on the contribution of dynamic modelling. Let us rewrite Euler equation (3), at the symmetric equilibrium as

$$\varepsilon(e_t) = V_1(e_t, X_t, e_t) - \underbrace{\beta((1-\delta)V_1(e_{t+1}, X_{t+1}, e_{t+1}) - V_2(e_{t+1}, X_{t+1}, e_{t+1}))}_{h_{t+1}} \quad (10)$$

Taking as given expectations, and henceforth the term h_{t+1} , and as X_t is predetermined, the period- t temporary equilibrium is the effort e_t that solves $\varepsilon(e_t) = 0$. We assume that at least one period- t temporary equilibrium exists in Assumption A4⁴. A sufficient condition for uniqueness is $\varepsilon'(e) < 0$, which is also part of Assumption A4. The condition for uniqueness is that complementarities should not be too strong.

⁴Here again, one could derive sufficient conditions for the existence of at least one period- t temporary equilibrium, but such conditions are not particularly interesting.

Assumption A4.

(i) *At least one period- t equilibrium exists,*

(ii) $V_{13} < \tilde{V}_{13} \equiv -\theta$.

In the local dynamics, we will have $\gamma_0 < 1$ if and only if Assumption A4 holds. In symmetric equilibrium, assuming $\alpha_0 + \gamma_1 \neq 0$, we have the bivariate system

$$\begin{pmatrix} \hat{X}_{t+1} \\ \hat{e}_{t+1} \end{pmatrix} = \underbrace{\begin{pmatrix} (1-\delta) & 1 \\ \phi_0 & \phi_1 \end{pmatrix}}_A \begin{pmatrix} \hat{X}_t \\ \hat{e}_t \end{pmatrix}$$

where

$$\begin{aligned} \phi_0 &\equiv -\frac{\alpha_1}{\alpha_0 + \gamma_1} = \frac{\sigma}{\beta [\chi + (1-\delta) V_{13} - V_{23}]} \\ \phi_1 &\equiv \frac{1 - \gamma_0}{\alpha_0 + \gamma_1} = \frac{\theta + V_{13}}{\beta [\chi + (1-\delta) V_{13} - V_{23}]} \end{aligned}$$

The eigenvalues of this system are

$$\begin{aligned} \lambda_0 &= \frac{(1 - \delta + \phi_1) - \sqrt{(1 - \delta - \phi_1)^2 + 4\phi_0}}{2} \\ \lambda_1 &= \frac{(1 - \delta + \phi_1) + \sqrt{(1 - \delta - \phi_1)^2 + 4\phi_0}}{2} \end{aligned}$$

Note that the roots are real if and only if $(1 - \delta - \phi_1)^2 + 4\phi_0 \geq 0$. It will also be useful to define the characteristic polynomial (of which these eigenvalues are the roots) by

$$P(\lambda) = \lambda^2 - (1 - \delta + \phi_1) \lambda + (1 - \delta) \phi_1 - \phi_0$$

Note that P is convex, and thus λ_0, λ_1 are real if and only if there exists a value of λ for which $P(\lambda) \leq 0$. Further, if $P(\lambda) < 0$ for some λ , then $\lambda_0 < \lambda < \lambda_1$. This will be a useful result later on.

Finally note that if $\alpha_0 + \gamma_1 = 0$ (i.e., if $(1 - \delta) V_{13} - V_{23} = -\chi$), then we lose the forward-looking element, and the system collapses to the univariate one

$$\hat{X}_{t+1} = \frac{V_{23}}{\theta + V_{13}} \hat{X}_t \tag{11}$$

1.4 Equilibrium steady state

Here we simply recapitulate the steady state results previously obtained.

Proposition 1. *(i) Under Assumption A2, the steady state exists and is unique absent of interactions.*

(ii) Under Assumption A2, the steady state exists and is unique with substitutabilities (when $\psi < 0$).

(iii) Under Assumptions A2 and A3, the steady state exists and is unique if complementarities are not too strong ($\psi(X^s) < |\kappa(X^s)|$).

(iv) Under Assumptions A2 and A3, there will two steady states appearing below and above the no-interactions one if complementarities are strong enough ($\psi(X^s) > |\kappa(X^s)|$).

See Appendix for proofs.

2 Some Easy Cases to Grasp Intuition

In order to give intuitions for our general results, we specialize the model so that we can fully derive an order one dynamics and illustrate the type of emergent properties we can obtain, namely hysteresis, sunspots equilibrium or limit cycles, depending whether we will have fold or flip bifurcations. We consider two specific sets of restrictions on the model fundamentals. First, we assume $\beta = 0$, and we show that when V_{13} is increased starting from zero, hysteresis will occur if $V_{12} < 0$ and limit cycles if $V_{12} > 0$. Second, we assume $\beta > 0$, $\delta = 1$ and $V_{12} = 0$, and we show that we will have indeterminacy and therefore sunspot equilibria when V_{23} is increased starting from zero.

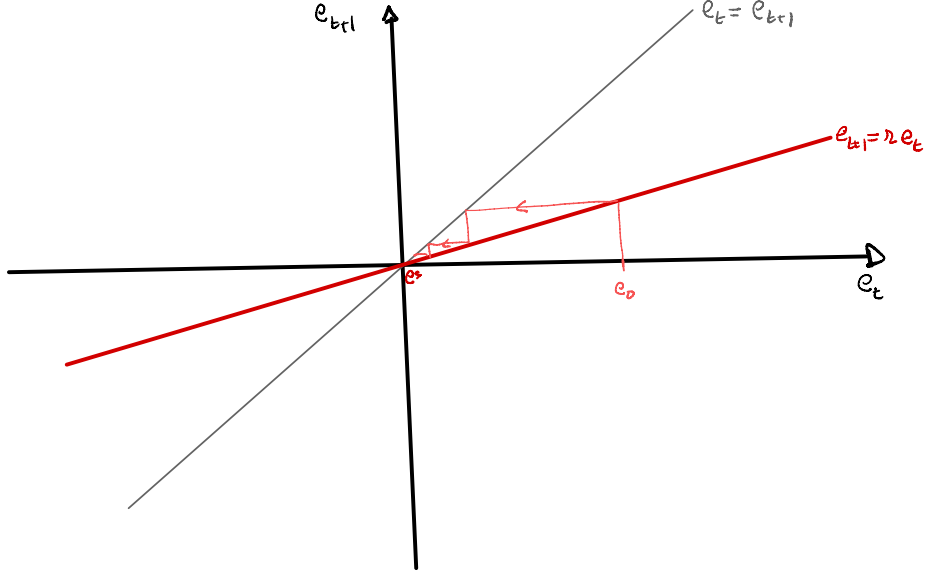
In that section, we assume that fundamentals are such that, absent of interactions, the steady state exists, is unique, that the equilibrium is determinate and that the convergence is non-oscillating.

2.1 A Case for Hysteresis and Limit Cycles

Here we assume $\beta = 0$. In such a case, equation (7) rewrites

$$\hat{e}_t = \frac{-V_{12}}{V_{11} + V_{13}} \hat{X}_t, \quad (12)$$

Figure 1: Non-oscillating convergent dynamics when $V_{13} = 0$



Notes: this figure depicts the model dynamics when $\beta = 0$ and $V_{13} = 0$. In that case, e_t is predetermined and there is non oscillating convergence to the steady state.

and using the accumulation equation (1), we obtain the order one linear dynamics around the steady state:

$$\hat{e}_{t+1} = \underbrace{\left(1 - \delta - \frac{V_{12}}{V_{11} + V_{13}}\right)}_r \hat{e}_t. \quad (13)$$

Note that equation (12) implies that e_t is predetermined, so that the model solution is fully backward, which is an obvious result given that $\beta = 0$. We assume that there is non-oscillating convergence without interactions – ie when $V_{13} = 0$, which is guaranteed if and only if $0 \leq r < 1$, which gives the condition $V_{12} \in [(1 - \delta)V_{11}, -\delta V_{11}[$. The dynamic absent of interactions is pictured in Figure 1.

Let us now introduce interactions by assuming that $V_{13} \neq 0$. There will be substitutabilities (respectively complementarities) between actions when $V_{13} < 0$ (respectively $V_{13} > 0$), as the best response of agent i to an increase in aggregate effort e_t will be to decrease (respectively to increase) e_{it} . A few interesting results can be stated looking at the function $r(V_{13})$, where r is the common ratio of the approximated order one equilibrium dynamics.

Results will depend on the sign of V_{12} , that represents the effect of the individual stock of past cumulated actions on the current individual level of effort. The sign of V_{12} affects the type of bifurcation that will occur, and this result extends to our more general configurations. When $V_{12} > 0$, high individual accumulation X_{it} makes current effort e_i (and then future accumulation) higher. That would be the case if we interpret X_{it} as productive capital that is combined with e_{it} to produce a consumption good, provided that the wealth effect of owning more X_{it} is not too strong compared to the substitution effect implied by the increased productivity of e_{it} . With our notations, all those effects are subsumed in the V function, and henceforth $V_{12} > 0$ corresponds to a case in which the substitution effect dominates the wealth effect. In the last section, we will give explicit examples in which we model utility, production and interactions and derive the implied V function. When $V_{12} < 0$, high accumulation X_{it} makes current effort e_{it} (and then future accumulation) lower. That would be the case if we interpret X_{it} as a stock of durable good and e as the production effort of new durable goods, as long as the marginal utility of durable goods is decreasing. It will also be the case if X_{it} is productive capital but the wealth effect on effort supply is strong.

The first interesting result that we can state is that, when introducing substitutabilities ($V_{13} < 0$), r stays in $[0, 1]$, so that the qualitative dynamics is not affected (non-oscillating convergence).

Things are different with complementarities. First, as V_{13} increases from zero, $|r|$ always exits the unit interval for some V_{13} large enough (close enough to $-V_{11}$), so that we have a bifurcation. The nature of the dynamics at the bifurcation depends on the sign of V_{12} . If $V_{12} > 0$, r stays positive and crosses one from below. The steady state becomes locally unstable (panel (a) of Fig 2). This is known as a fold bifurcation. At the bifurcation, two other locally attractive steady states emerge (one above and one below e^s). This can be seen in panel (b) of Figure 2, where we look at the global dynamics of the economy. As it can be seen, the dynamics displays hysteresis. Depending on which side of the no-interactions steady state it starts, the economy converges towards one of the two “other” steady states (panel (c) of Figure 2). If $V_{12} < 0$, r becomes negative at some point. The steady state is still stable, but the convergence is oscillating (panel (d) of Figure 2). Then, when V_{13} continues

increasing, r stays negative and crosses -1 from above. The steady state then becomes locally unstable (panel (e) of Figure 2). This is known as a flip bifurcation. At the bifurcation, a limit two-cycle emerges around the no-interactions steady state e^s . The economy cycles absent of any shocks (panel (f) of Figure 2).

2.2 A Case for Sunspot Equilibria

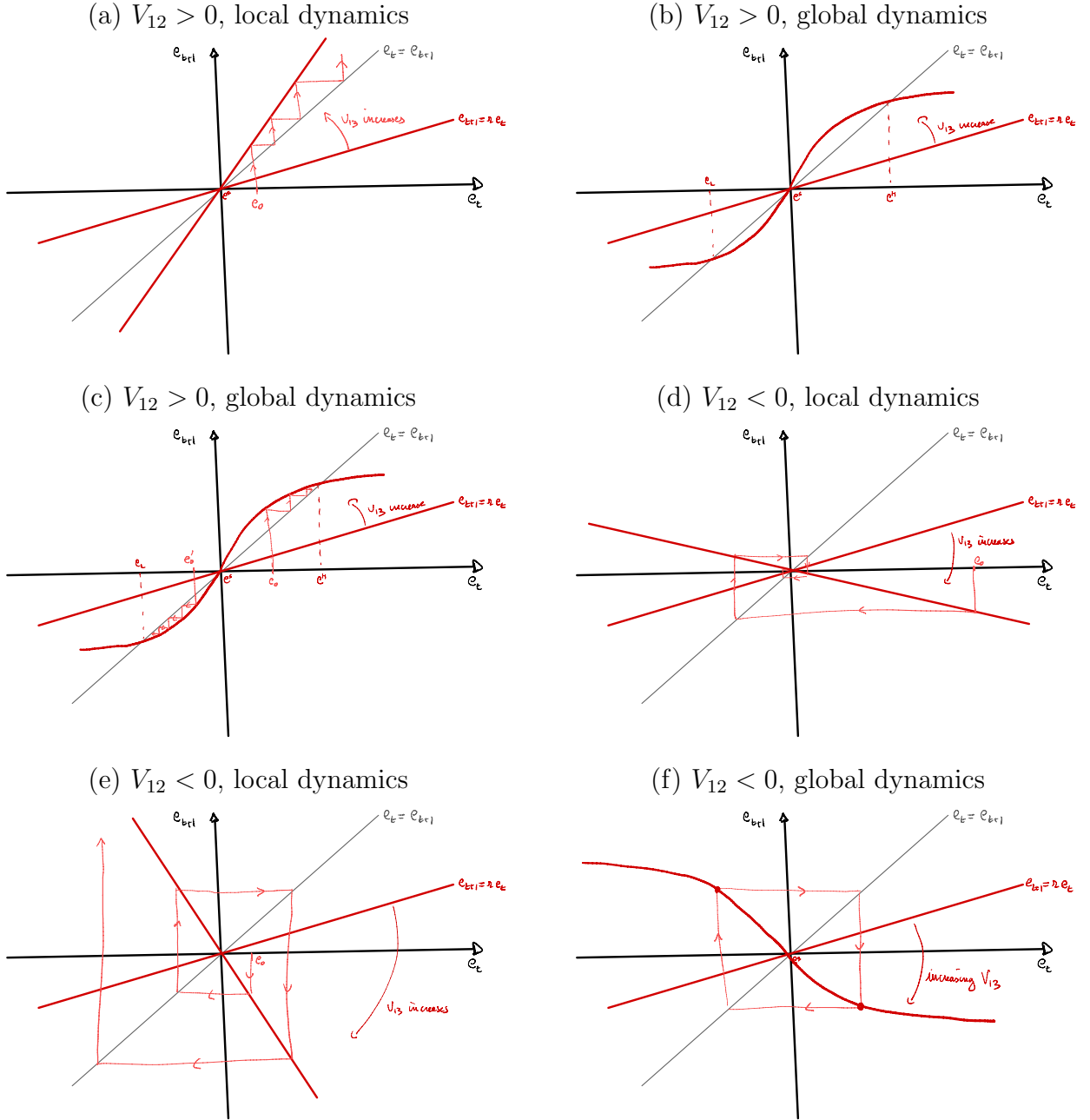
Here we consider another extreme configuration of our general setup in which the dynamics is of order one, namely $\delta = 1$ and $V_{12} = V_{13} = 0$. In this case, equations (7) and (1) imply

$$\hat{e}_t = - \underbrace{\frac{\beta V_{23}}{V_{11} + \beta V_{22}}}_{1/r} \hat{e}_{t+1}. \quad (14)$$

Interactions are here indexed by V_{23} , which represents the effect of aggregate effort on the value of the individual stock X_{it} . $V_{23} > 0$ would for example correspond to an economy in which high level of aggregate activity e_t today pushes up the demand for the services of X_{it} , and henceforth its value V_2 . $V_{23} < 0$ would correspond to an economy in which high current aggregate activity creates congestion that reduces the current value of individual stock.

We start with the no-interactions case, for which $V_{23} = 0$. The model has no dynamics and $e_t = 0 \forall t$. We now introduce interactions by increasing $|V_{23}|$. If $|V_{23}|$ is not too large (the exact condition is $|V_{23}| < |V_{11} + \beta V_{22}|$), $|1/r|$ stays below one. The equilibrium stays determinate, and the economy always jumps to the no-interaction steady state (panel (a) and (b) of Figure 3). Increase further interactions (positive or negative) will cause $|1/r|$ to pass above one, and the economy loses determinacy. For an initial and common X_{i0} , *any* choice e_{i0} implies an approximated equilibrium dynamics $\hat{e}_{t+1} = r \hat{e}_t$ that satisfies the transversality condition, and is therefore an equilibrium solution. The equilibrium path is indeterminate, as any extraneous signal (a sunspot) that coordinate agents on an arbitrary effort $e_{i0} = e_0$ is an equilibrium. With substitutabilities, $r > 0$ so that the equilibrium dynamics is non-oscillating (panel (c) of Figure 3), while cycles occur with complementarities (panel (d) of Figure 3).

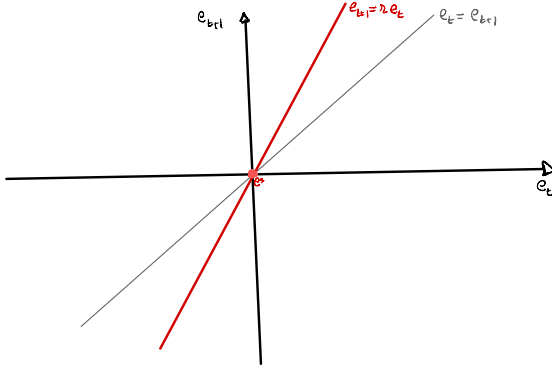
Figure 2: Fold or Flip Bifurcations when V_{13} Increases



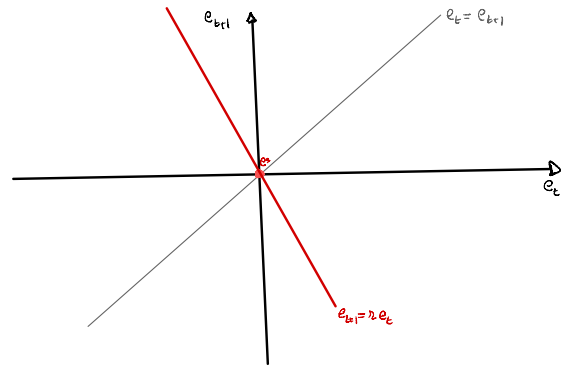
Notes: the panels of this figure depict the model dynamics when $\beta = 0$ and $V_{13} > 0$, for V_{12} positive or negative. In all cases, e_t is predetermined.

Figure 3: Indeterminacy with or without cycles when $|V_{23}|$ Increases

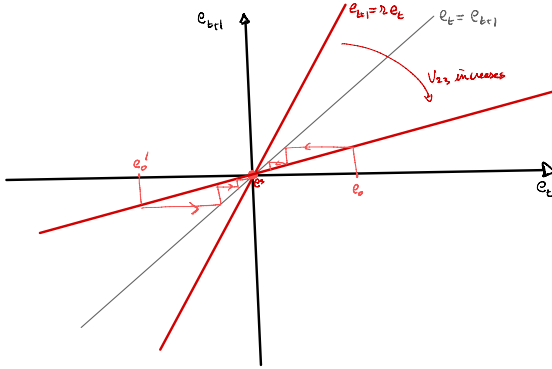
(a) $-(V_{11} + \beta V_{22}) > V_{23} > 0$, local dynamics



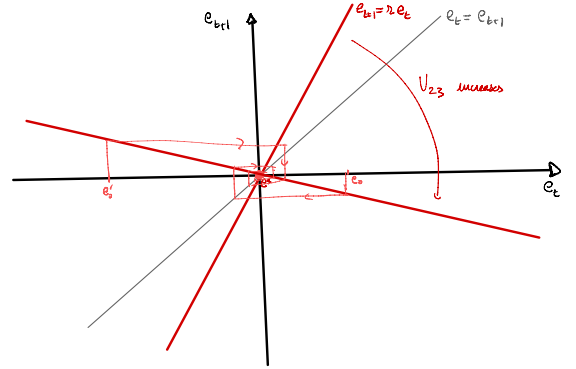
(b) $(V_{11} + \beta V_{22}) < V_{23} < 0$, local dynamics



(c) $V_{23} > -(V_{11} + \beta V_{22})$, local dynamics



(d) $V_{23} < (V_{11} + \beta V_{22})$, local dynamics



Notes: the panels of this figure depict the model dynamics when $\delta = 1$ and $V_{12} = V_{13} = 0$, for V_{12} positive or negative. In all cases, e_t is not predetermined.

2.3 Summary

From those simple cases, we can infer some “intuitions” that will prove to be correct in a more general setup. First, complementarities between actions favour determinate hysteresis and cycles, substitutabilities between actions favour determinacy and stability. Second, interactions between aggregate action and individual stock (complementarities or substitutabilities) favour indeterminacy and hence sunspot equilibria. Third, in all the cases, whether current stock marginally increases or decreases incentives to accumulate is the key parameter to determine which type of bifurcation will occur or whether the dynamics will be oscillating or not.

3 More General Results

Now we consider some more general cases of our model, in which we will confirm that the “intuitions” obtained from the order one dynamics examples are indeed correct. We study successively cases in which either $V_{23} = 0$ or $V_{13} = 0$, and can derive relatively nice and interpretable results. Then, we give some micro-fundations for those specific cases.

3.1 Dynamics in the No-Interactions Case

Consider the no-interactions case, $V_{13} = V_{23} = 0$, so that $\gamma_0 + \gamma_1 = 0$. Note that in this case $\alpha_0 + \gamma_1 = \alpha_0 > 0$, and thus we are always in the bivariate system case. We have the following proposition.

Proposition 2. *Suppose [A1](#) and [A2](#) hold, and $V_{13} = V_{23} = 0$. Then the eigenvalues are real, with $0 < \lambda_0 < 1 < \lambda_1$.*

Proposition [2](#) establishes that, with no interactions, under Assumptions [A1](#) and [A2](#) are sufficient to ensure that there is a unique (determinate) solution involving non-oscillating convergence to the steady state. We now study the implications of introducing interactions ($V_{13} \neq 0$ or $V_{23} \neq 0$).

3.2 Adding Interactions: e only

We suppose that $V_{23} = 0$, so that the interaction only comes through e_t . It will be useful to define

$$\bar{V}_{13} = -\frac{1}{1-\delta}\chi > 0$$

as the value of V_{13} at which the system collapses to the univariate process. In Proposition 3, we show what happens up to the first bifurcation⁵ when V_{13} is increased or decreased away from zero.

Proposition 3. *Suppose A1 and A2 hold, and $V_{23} = 0$.*

1. *For any $V_{13} \leq 0$, we have real eigenvalues with $0 < \lambda_0 < 1 < \lambda_1$ (unique solution, non-oscillating convergence).*
2. *If $\sigma = 0$, then the eigenvalues are unaffected by V_{13} , and thus we always have $0 < \lambda_0 < 1 < \lambda_1$.*
3. *If $\sigma < 0$, then beginning from $V_{13} = 0$, as V_{13} increases we observe the following behavior (in order):*
 - (a) *Initially we have real eigenvalues with $0 < \lambda_0 < 1 < \lambda_1$ (unique solution, non-oscillating convergence).*
 - (b) *This structure continues until $V_{13} = \bar{V}_{13}$, at which point the system temporarily loses a dimension, and the lone eigenvalue is zero (no dynamics).*
 - (c) *As V_{13} increases slightly above $\bar{V}_{13}$, the system regains the second dimension, and we have real eigenvalues with $\lambda_0 < -1 < \lambda_1 < 0$ (unique solution, damped oscillation).*
 - (d) *This structure continues until V_{13} increases through a point V_{13}^* , at which point λ_1 decreases through -1 and the system loses local stability, though the roots are still real-negative with $\lambda_0 < \lambda_1$.*
4. *If $\sigma > 0$, then beginning from $V_{13} = 0$, as V_{13} increases we observe the following behavior (in order):*

⁵We can actually characterize what happens after the first bifurcation, but leave it aside for clarity of exposition. Results are available upon request.

- (a) Initially we have real eigenvalues with $0 < \lambda_0 < 1 < \lambda_1$ (unique solution, non-oscillating convergence).
 - (b) This structure continues until V_{13} increases through a point V'_{13} , at which point λ_0 increases through one and the system loses local stability (both real, positive, unstable).
5. When $\sigma \neq 0$, the point $V_{13} = \tilde{V}_{13}$ occurs in the region of complex roots with positive real part.

The results stated in Proposition 3 are well in line with the results we obtained in the dimension-one case. When one introduces substitutabilities, the dynamics is not qualitatively affected: the equilibrium is determinate and the transition to the steady state is non-oscillating.

With complementarities, the type of bifurcation occurring will depend on the sign of $\sigma = [1 - \beta(1 - \delta)^2] V_{12} + \beta(1 - \delta) V_{22}$. σ will be positive only if V_{12} is positive, but will be always negative if V_{12} is negative. As already mentioned, when V is sufficiently concave in X_{it} , and hence V_{22} negative and sufficiently large in absolute value, it is possible to have $\sigma > 0$ even though V_{12} is positive.

Consider first what happens when $\sigma < 0$ (which is almost the case $V_{12} < 0$, as explained in the previous paragraph). We refer to this case as a case with “durability”, in the sense that, as for a durable good, the higher is the stock, the lower the new purchases. In this case, increasing enough complementarities will first keep the steady state determinate, but will make the convergence oscillating. Cycles are indeed a natural outcome in such a case. Pushing further the level of complementarities, but staying below the level that would imply indeterminacy of the period- t equilibrium, the stable eigenvalue of the linearized system will pass -1, so that we will have a flip bifurcation, and hence a limit cycle of order two. In the static world Cooper and John [1988], complementarities were the foundations of multiple equilibria; here they are a foundation of self-sustained fluctuations in a dynamic set up with “durability”.

When $\sigma > 0$, we must have $V_{12} > 0$, and we refer to that case as a “capital” case, as in a model in which having more capital is marginally increasing incentives to work.⁶ In such

⁶This is true if the wealth effect is not too strong. Also note that a model with habit persistence in which

a case, increasing complementarities in actions will eventually lead to a fold bifurcation, when the stable eigenvalue will cross 1. At the bifurcation, the initial steady state becomes unstable and two determinate steady state with non oscillating convergence will emerge, one above and one below the no-interactions one. We will then have hysteresis, as the economy will be determinate and converge to one of the two new steady states depending on its initial condition. Note that those two high and low steady states are determinate: it is not coordination of expectation but initial conditions that select the trajectory converging towards one of the two.

3.3 Adding Interaction: X only

Let us now suppose that $V_{13} = 0$ and allow $V_{23} \neq 0$. It will be useful to define the following quantity:

$$\hat{\sigma} \equiv \delta^2 V_{11} + [1 - \beta(1 - \delta) + \delta] V_{12} + \beta V_{22}$$

In Proposition 4, we show what happens up to the first bifurcation when V_{23} is increased or decreased away from zero.

Proposition 4. *Suppose A1 and A2 hold, and $V_{13} = 0$.*

1. *Suppose $\hat{\sigma} < 0$. Then as V_{23} increases from zero, we observe the following behavior:*
 - (a) *Initially we have real eigenvalues with $0 < \lambda_0 < 1 < \lambda_1$ (unique solution, non-oscillating convergence).*
 - (b) *The first transition occurs at a value $V_{23} = V_{23}^* > 0$ at which λ_1 decreases through one, so that we have $0 < \lambda_0 < \lambda_1 < 1$ (indeterminacy).*
2. *Suppose $\hat{\sigma} > 0$. Then as V_{23} increases from zero, we observe the following behavior (in order):*
 - (a) *Initially we have real eigenvalues with $0 < \lambda_0 < 1 < \lambda_1$ (unique solution, non-oscillating convergence).*
 - (b) *This structure continues until V_{23} increases through some value V_{23}^* at which point λ_0 increases through one, so that we have $1 < \lambda_0 < \lambda_1$ (local instability emerges).*

X will stand for the stock of habit will also deliver a positive V_{12} .

3. As V_{23} decreases from zero, we observe the following behavior (regardless of the value of $\hat{\sigma}$):

- (a) Initially we have real eigenvalues with $0 < \lambda_0 < 1 < \lambda_1$ (unique solution, non-oscillating convergence).
- (b) This structure continues until $V_{23} = \chi < 0$, at which point the system temporarily loses a dimension, and the lone eigenvalue is $\chi/\theta \in (0, 1)$ (unique solution, non-oscillating convergence).
- (c) As V_{23} decreases slightly below χ , the system regains the second dimension, and we have real eigenvalues with $\lambda_0 < -1$, $0 < \lambda_1 < 1$ (unique solution, non-oscillating convergence).
- (d) This continues until V_{23} decreases through a point V_{23}^{***} , at which point λ_0 increases through -1 , so that we have $-1 < \lambda_0 < 0 < \lambda_1 < 1$ (indeterminacy).

Again, the results stated in Proposition 4 are well in line with the results of the previous section. The sign of $\hat{\sigma} = \delta^2 V_{11} + [1 - \beta(1 - \delta) + \delta] V_{12} + \beta V_{22}$ is important for the type of bifurcation that occurs. In the “durable” case, V_{12} is negative so that $\hat{\sigma}$ is negative. In the “capital” case, if V_{12} is positive enough, then $\hat{\sigma}$ can be positive. Note that if the model’s period is short (say a quarter), δ will be close to 0, β close to 1 so that $1 - \beta(1 - \delta) + \delta$ is close to zero. In such a case, it will need high V_{12} for $\hat{\sigma}$ to be positive.

With substitutabilities, regardless of the sign of $\hat{\sigma}$, the bifurcation will be an indeterminacy-inducing fold one, with the unstable eigenvalue entering the unit interval. We will then have indeterminacy and sunspot equilibria.

With complementarities, the sign of $\hat{\sigma}$ (and therefore the value of V_{12}) will matter. In the “durable” case, we will have indeterminacy-inducing fold bifurcation whereas in the “capital” case, if V_{12} is large enough, we will have a hysteresis-inducing fold bifurcation.

3.4 Microfoundations of the V function: a set of examples

As an illustration, we give here some examples of general equilibrium models in which a V function of the type $V(e_{it}, X_{it}, e_t)$ can be derived, satisfies either $V_{23} = 0$ or $V_{13} = 0$, and in which one can sign V_{12} .

3.4.1 Productive externalities with habit persistence

There is a continuum of measure one of household (i) and firms (j). Suppose preferences of household i are given by

$$\sum_{t=0}^{\infty} \beta^t \left(\log(c_{it} - \gamma h_{it}) - \frac{\omega_0}{\omega} \ell_{it}^{1+\omega} \right)$$

where the stock of habit h accumulates according to $h_{it+1} = (1 - \delta)h_{it} + c_{it}$. All parameters are positive. Production function of firm j is

$$c_{jt} = \chi_t \ell_{jt}^{\alpha},$$

and interactions are modelled as productive externalities according to

$$\chi_t = \chi \ell_t^{\nu\alpha},$$

where ℓ_t is the average labour effort in the economy and parameter ν sets the strength of interactions. Aggregating over j gives

$$\ell_t = \left(\frac{c_t}{\chi} \right)^{\frac{1}{\alpha(1+\nu)}}.$$

As the labor market is competitive, wage w_t will equal private marginal productivity of labor, such that

$$w_t = \alpha \chi \ell_t^{\frac{1}{\alpha(1+\nu)} - 1}.$$

The budget constraint of household i is

$$c_{it} = w_{it} \ell_{it} + \pi_t,$$

assuming that profits are equally redistributed to households. In equilibrium, using the expression for the wage, the budget constraint will imply

$$c_{it} = \alpha \chi \ell_t^{\frac{1}{\alpha(1+\nu)} - 1} \ell_{it} + (1 - \alpha) c_t,$$

or equivalently

$$\ell_{it} = \frac{1}{\alpha \chi} \left(c_{it} - (1 - \alpha) c_t \right) \left(\frac{c_t}{\chi} \right)^{\frac{1 - \alpha(1+\nu)}{\alpha(1+\nu)}}.$$

Denote $e_{it} = c_{it}$ and $X_{it} = h_{it}$, instantaneous utility can be rewritten as

$$V(e_{it}, X_{it}, e_t) = \log \left(e_{it} - \gamma X_{it} \right) - \frac{\omega_0}{1 + \omega} \alpha^{-(1+\omega)} \chi^{\frac{(1+\omega)}{\alpha(1+\nu)}} \left(e_{it} - (1 - \alpha e_t) \right) e_t^{(1+\omega) \left(\frac{1}{\alpha(1+\nu)} \right) - 1}.$$

It is straightforward to check that $V_{13} \neq 0$, $V_{23} = 0$ and $V_{12} > 0$. In this model, the bifurcation parameter, that will directly determine V_{13} , will be the level of productive externalities parameter ν .

3.4.2 Productive externalities with durable good

Suppose preferences of household i are instead given by

$$\sum_{t=0}^{\infty} \beta^t \left(\log(c_{it} + \gamma d_{it}) - \frac{\omega_0}{\omega} \ell_{it}^{1+\omega} \right)$$

where the stock of durables and d accumulates according to $d_{it+1} = (1 - \delta)d_{it} + c_{it}$. The rest of the model is the same than before. Denoting $X_{it} = d_{it}$ and $e_{it} = c_{it}$, instantaneous utility can be rewritten as

$$V(e_{it}, X_{it}, e_t) = \log(e_{it} + \gamma X_{it}) - \frac{\omega_0}{1 + \omega} \alpha^{-(1+\omega)} \chi^{\frac{(1+\omega)}{\alpha(1+\nu)}} \left(e_{it} - (1 - \alpha e_t) e_t^{(1+\omega)(\frac{1}{\alpha(1+\nu)})-1} \right).$$

It is straightforward to check that $V_{13} \neq 0$, $V_{23} = 0$ and $V_{12} < 0$. Again, the bifurcation parameter will be the level of productive externalities parameter ν .

3.4.3 Habit persistence meets “keeping up with the Joneses”

Suppose preferences of household i are given by

$$\sum_{t=0}^{\infty} \beta^t \left(c_{it} - \gamma h_{it} c_t^\nu - \frac{\omega_0}{\omega} \ell_{it}^{1+\omega} \right)$$

where the stock of habits h accumulates according to $h_{it+1} = (1 - \delta)h_{it} + c_{it}$ and where consumption good is produced according to $c_{it} = \chi \ell_{it}$. In this habit persistence model, the reference point is a function of both cumulated past consumption h_{it} and current consumption of the others c_t . Denoting $X_{it} = h_{it}$ and $e_{it} = c_{it}$, instantaneous utility can be rewritten as

$$V(e_{it}, X_{it}, e_t) = \left(e_{it} - \gamma X_{it} e_t^\nu \right) - \frac{\omega_0}{1 + \omega} \chi^{1+\omega} e_{it}^{1+\omega}.$$

It is straightforward to check that $V_{13} = 0$, $V_{23} \neq 0$ and $V_{12} < 0$. In this model, the bifurcation parameter, that will directly determine V_{23} , will be the level of the “keeping up with the Joneses” parameter ν .

4 Beyond the Tractable Cases: Numerical Examples

In this section, we want to provide some numerical examples showing that the results we have obtained in our admittedly restricted setup seem to be useful for larger models.

4.1 Model

Let us write a model that embeds the features introduced in the previous section, namely habit persistence, durability, keeping up with the Jonesese and productive externality.

Preferences are given by

$$\sum_{t=0}^{\infty} \beta^t \left(\log \left(\psi d_{it} + c_{it} - (1 - \psi) \gamma h_{it}^{\mu} c_t^{1-\mu} + \bar{c} \right) - \frac{\omega_0}{1 + \omega} \ell_{it}^{1+\omega} \right),$$

accumulation follows

$$\begin{aligned} d_{it+1} &= (1 - \delta_d) d_{it} + \gamma_d c_{it}, \\ h_{it+1} &= (1 - \delta_h) h_{it} + \gamma_h c_{it}, \end{aligned}$$

where d_{it} is the stock of durable goods and h_{it} the stock of habit. Firm j produces according to

$$c_{jt} = \chi_t \ell_{jt}^{\alpha}.$$

Externalities in production are modelled as

$$\chi_t = \chi \ell_t^{\nu\alpha}.$$

In the analysis, ν is the bifurcation parameter, and we will show that depending on the other parameter values, different type of bifurcations will happen when ν is increased. In this model, the V function does not have only three arguments, and writes $V(c_{it}, h_{it}, d_{it}, c_t, h_t, d_t)$.

4.2 Three configurations for parameters

Parameters are chosen to be “in the range of admissible values”, without any calibration strategy. The parameters in Table 1 are the same in all configurations. We choose ω_0 such that that the steady level of hours is one when $\nu = 0$ in each configuration. The parameters in Table 2 are specific to each case, so that we can contrast different situations.

Table 1: Common Parameters Values

β	α	χ	ω	γ_d	γ_h	$\bar{c}$	δ_d
.98	.7	1	.5	.5	.5	1	.025

Notes: ω_0 is set such that steady state level of hours is one when $\nu = 0$ in all models.

Table 2: Specific Parameters Values

	ψ	γ	μ	δ_h
Case I: Limit cycle	.5	1	1	.9
Case II: Hysteresis	.3	1	1	.05
Case III: Sunspots	.1	1	.7	.025

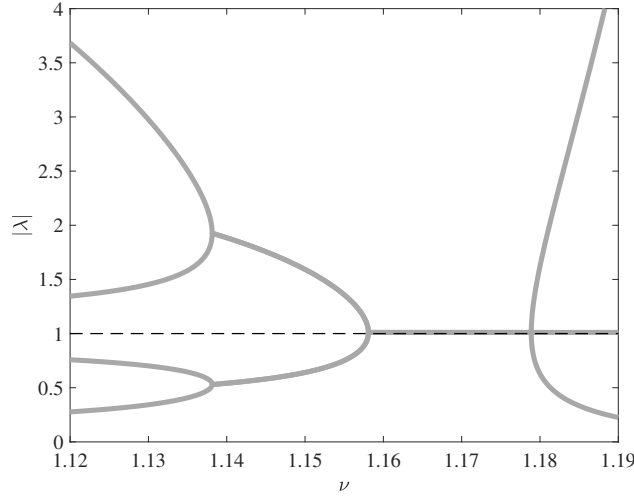
The model dimension is four, so that we have four eigenvalues. In the benchmark case in which $\nu = 0$, we have determinate and non oscillating dynamics, so that two eigenvalues are inside the unit disk (real and positive) and two outside.

We study three configurations of parameters. The first configuration (case I) is one in which durability is strong, and in which the habit stock depreciates strongly. We shall be in that case close to the Flip bifurcation case of the two dimensional theoretical analysis, and we might expect limit cycles to occur. The second configuration (case II) is one in which habit persistence is strong, which we expect to produce hysteresis-inducing fold bifurcation. In the third configuration (case III), habit persistence dominates and it is partly external, which we expect to produce sunspots-inducing fold bifurcation.

In Figures 4, 5 and 6, we plot for each case the moduli of the four eigenvalues and Figure 7 shows the values of the non-zero derivatives of the V function at $\nu = 0$ and at the first bifurcation.

Consider Figure 4, that corresponds to case I. From $\nu = 0$ to $\nu = 1.12$, there are four real distinct eigenvalues, two larger than one in modulus, and two smaller, so that the equilibrium is determinate. complementarities further increase, two pairs of complex eigenvalues emerge: the equilibrium becomes cyclical, although still determinate and converging to the steady state. It can be further checked that all the eigenvalues are positive, so that there are no

Figure 4: Case I: Hopf Bifurcation when ν is Increased (Limit cycle case)

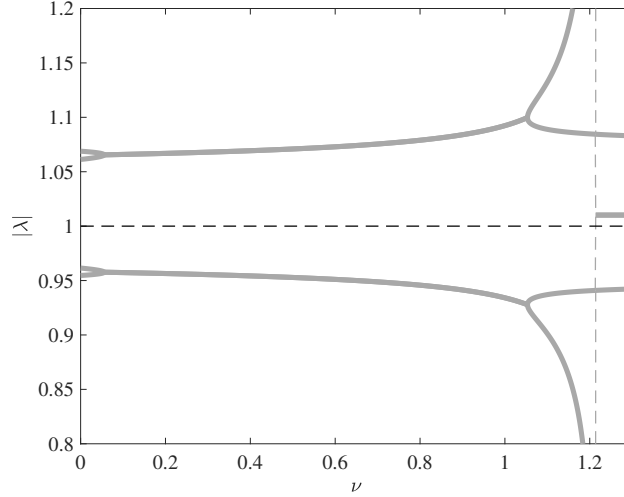


Notes: This Figure plots the modulus of the model four eigenvalues when ν is increased. When a pair of complex conjugate eigenvalues emerges, their two modulus are equal. From $\nu = 0$ to $\nu = 1.12$, there are four real distinct eigenvalues, two larger than one in modulus, and two smaller.

oscillations. Around $\nu = 1.16$, a bifurcation occurs: the pair of eigenvalues inside the unit disk exits, which is a Hopf bifurcation. There, a smooth limit cycle occurs. Increasing further ν , the four eigenvalues collapse to the same modulus, slightly larger than one. The next bifurcation occurs when one eigenvalue enters again the unit disk while the three other stay outside. The occurrence of a limit cycle is reminiscent of our theoretical results, although the cycle was then a 2-cycle (flip bifurcation) because the dynamical system was of dimension two, so that Hopf bifurcation was not possible starting from a determinate equilibrium when $\nu = 0$. Indeed, the inspection of the model V function derivatives (first column of Figure 7 show that we are in a similar case. When $\nu = 0$, $V_{c_i c}$ is negative, so that actions (here consumptions) are strategic substitute. Because the stock of habit h depreciates rapidly, the main stock variable is the stock of durables d , and one has $V_{c_i d_i} < 0$, which is reminiscent of the flip bifurcation configuration. Indeed, at the bifurcation, $V_{c_i c}$ is positive, $V_{c_i d_i}$ stays negative and we have a limit cycle ocuring.

Consider now Figure 5 and the second row of Figure 7, that corresponds to case II. Here, because ψ is small, the relevant stock if the stock of habit, and we have $V_{c_i h_i} > 0$. Increasing ν moves $V_{c_i c}$ from below to above zero, and $V_{c_i h_i}$ stays negative. This is a configuration

Figure 5: Case II: Hysteresis-Inducing Fold Bifurcation When ν is Increased

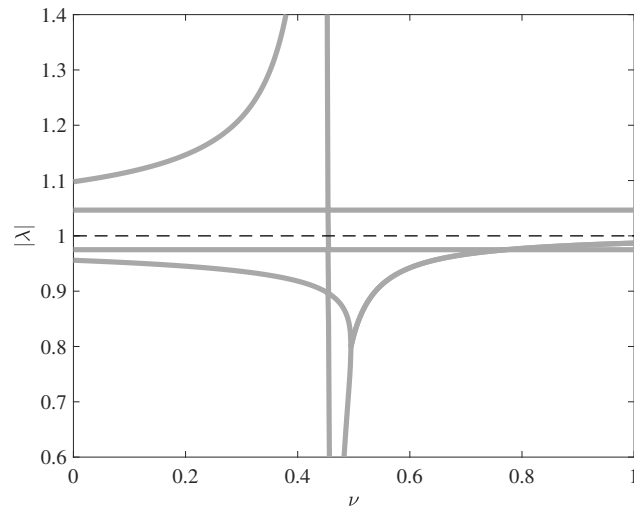


Notes: This Figure plots the modulus of the model four eigenvalues when ν is increased. When a pair of complex conjugate eigenvalues emerges, their two modulus are equal.

that corresponds to the hysteresis-inducing fold bifurcation in our analysis of simple two-dimensional dynamics. As shown in Figure 5, the economy starts with four distinct real eigenvalues, two above one and two below in modulus, so that the equilibrium is initially determinate. It can be further checked that all the eigenvalues are positive, so that there are no oscillations. Increasing ν , we first have the appearance of two pairs of complex eigenvalues: the equilibrium is still determinate but it becomes oscillating. Eigenvalues split again (four real ones), keeping determinacy. The first bifurcation occurs a bit before $\nu = 1.2$, where one of the eigenvalue with smaller than one modulus exit the unit disk, so that we have three eigenvalues become larger than one in modulus and one below one. That is an hysteresis-inducing fold bifurcation, as we had in the simple dimension-two dynamics.

Finally, consider Figure 6 and the third row of Figure 7, that corresponds to case III. Again, the equilibrium is determinate and non-oscillating when $\nu = 0$. Here again, because ψ is small, the relevant stock is the stock of habit, and we have $V_{c_i h_i} > 0$. Looking at the second order derivatives of V on Figure 7, we see that $V_{h_i c}$ is positive, while it was always zero in the previous cases. We are in a situation similar to the sunspot case of the previous section. Indeed, as shown in Figure 6, increasing ν to about .45, one explosive eigenvalue enters the unit disk, so that the equilibrium becomes indeterminate: the first bifurcation is

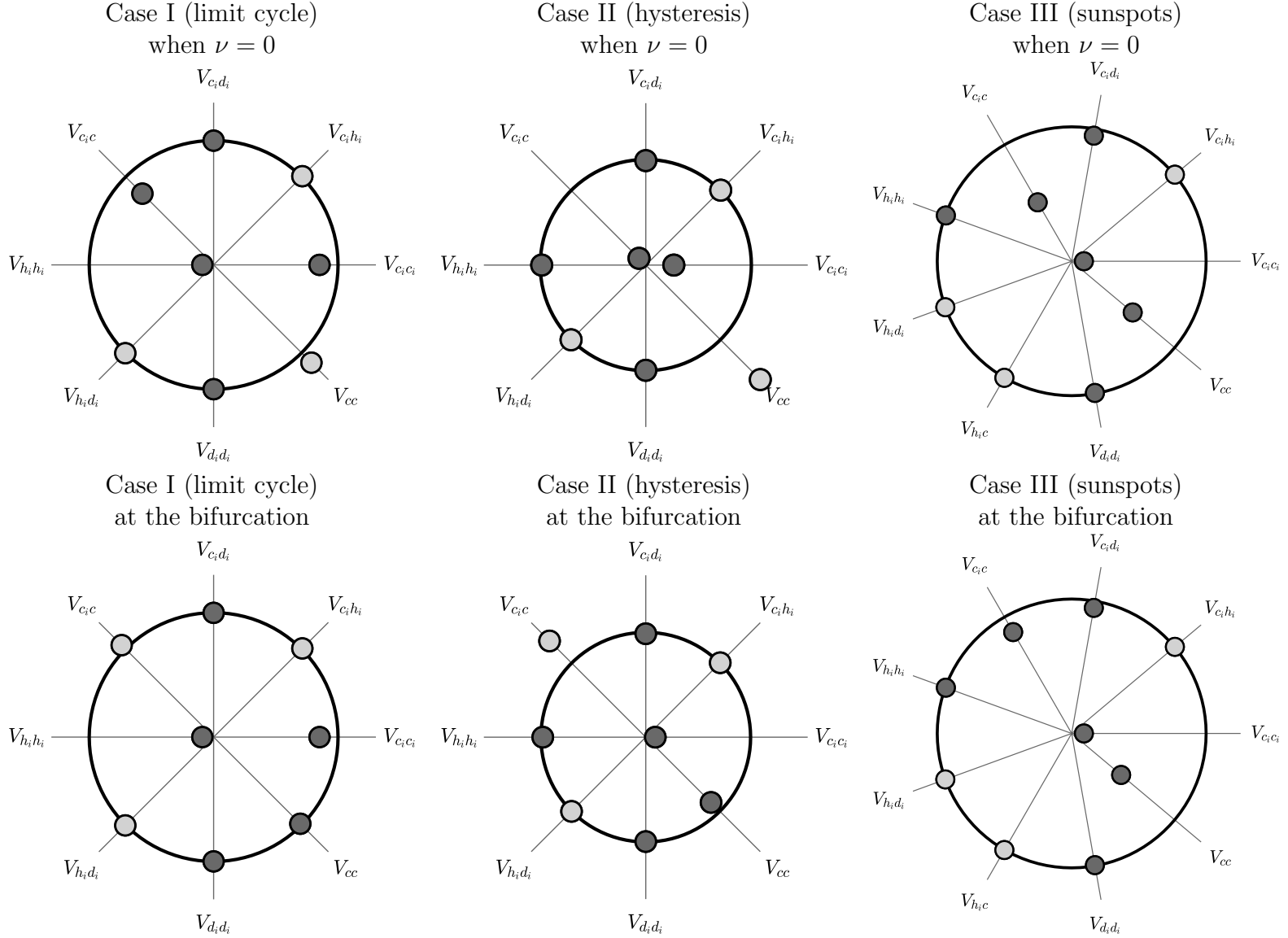
Figure 6: Case III: Sunspots-Inducing Fold Bifurcation When ν is Increased



Notes: This Figure plots the modulus of the model four eigenvalues when ν is increased. When a pair of complex conjugate eigenvalues emerges, their two modulus are equal.

indeed a sunspot-inducing fold bifurcation.

Figure 7: Non-Zero Second Derivatives of the V Function



Notes: Each panel plots the non-zero second derivative of the function $V(c_{it}, h_{it}, d_{it}, c_t, h_t, d_t)$, as obtained from the model of section 4. The black circle represents the value 0, while the position of each dot on each circle radius is the value of V_{ab} . A dark dot stands for a negative value (inside the circle) while a light dot indicates a positive value (outside the circle).

Conclusion

We have studied a model that can be seen as a dynamic extension of Cooper and John [1988]. In that framework, we have shown under which condition on the payoff structure does increasing complementarities between agents lead to emergent phenomena such as limit cycles, hysteresis or sunspot fluctuations.

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Appendix

A Proofs of Propositions

Preliminaries

The proofs make use of Figure 8, which shows the various configurations for the eigenvalues depending on the trace τ (horizontal dimension) and determinant Δ (vertical dimension) of the evolution matrix A . In particular, note that

$$\lambda_0 = \frac{\tau - \sqrt{\tau^2 - 4\Delta}}{2}$$
$$\lambda_1 = \frac{\tau + \sqrt{\tau^2 - 4\Delta}}{2}$$

Proof of Proposition 1

The proof of Proposition 1 follows directly from the developpements in subsection 1.3.

Proof of Proposition 2

We have

$$\phi_0 = \frac{\sigma}{\beta\chi}$$
$$\phi_1 = \frac{\theta}{\beta\chi}$$

Note that the product of the eigenvalues is $1/\beta > 1$, and thus if the eigenvalues are real then they are of the same sign. Next,

$$P(1) = -\frac{[1 - \beta(1 - \delta)]\delta V_{11} + [1 - \beta(1 - 2\delta)]V_{12} + \beta V_{22}}{\beta\chi} \quad (\text{A.1})$$

Note that by A2 the denominator of the fraction on the right-hand side of (A.1) is negative, while by A1 so is the numerator, and therefore $P(1) < 0$. This in turn implies that the roots are real with $\lambda_0 < 1$ and $\lambda_1 > 1$, and since the eigenvalues must be of the same sign, we must also have $\lambda_0 > 0$. which completes the proof.

Proof of Proposition 3

For $V_{13} = \bar{V}_{13}$, it is straightforward to see from (11) and the fact that $V_{23} = 0$ that part (2)(b) of the Proposition holds. Thus, henceforth consider only values $V_{13} \neq \bar{V}_{13}$. One can verify in this case that the determinant of the evolution matrix is $\Delta = 1/\beta$. Note first that this implies that neither of the roots can ever be zero, and that if they are real they must be of the same sign. Further, it implies that as V_{13} changes, it only affects the trace

$$\tau(V_{13}) = (1 - \delta) + \phi_1$$

Thus, in the diagram of Figure 8, changes in V_{13} cause movements along the horizontal line given by $\Delta = 1/\beta$. Further, as noted in Proposition 2, for the case where $V_{13} = 0$, we have $0 < \lambda_0 < 1 < \lambda_1$, which is in the region to the right of line A (and above the horizontal axis, as will always be the case). Next, we have

$$\tau'(V_{13}) = \frac{\partial \phi_1}{\partial V_{13}} = \frac{-\sigma}{\beta(1-\delta)^2(V_{13} - \bar{V}_{13})^2}$$

Thus, with the possible exception of at the lone discontinuity point at $V_{13} = \bar{V}_{13}$, τ is either strictly increasing in V_{13} everywhere (if $\sigma < 0$), strictly decreasing in V_{13} everywhere (if $\sigma > 0$), or constant everywhere (if $\sigma = 0$). If $\sigma = 0$, then we clearly have the same $0 < \lambda_0 < 1 < \lambda_1$ configuration regardless of the value of V_{13} . Thus, henceforth take $\sigma \neq 0$. Since $\bar{V}_{13} > 0$, this implies that $\tau(\cdot)$ is continuous and strictly monotonic on $V_{13} \leq 0$. If $\sigma > 0$, then τ increases as V_{13} decreases from 0, so that we forever remain in the region to the right of line A in Figure 8, i.e., we will always have $0 < \lambda_0 < 1 < \lambda_1$. If instead $\sigma < 0$, then τ decreases as V_{13} decreases from 0, and thus it is in principle possible for τ to cross the line A from the right. However, we have

$$\lim_{V_{13} \rightarrow -\infty} P(1) = -\frac{[1 - \beta(1 - \delta)]\delta}{\beta(1 - \delta)} < 0$$

which implies that, even for V_{13} arbitrarily large and negative, we still have roots satisfying $0 < \lambda_0 < 1 < \lambda_1$, i.e., line A is never crossed.

Next, note that

$$\phi_1 = \frac{\theta + V_{13}}{\beta[\chi + (1 - \delta)V_{13}]}$$

Thus, both the numerator and denominator of this expression are strictly increasing in V_{13} , without bound. Further, since $\chi, \theta < 0$, for $V_{13} = 0$ both numerator and denominator are negative. Thus, as V_{13} increases from 0, both the numerator and denominator eventually transition from negative to positive, each passing through 0 at exactly one point. Suppose we are at the point where the denominator passes through 0, i.e., we have $V_{13} = \bar{V}_{13}$. It can be verified that the numerator is given by

$$\theta + V_{13} = \frac{\sigma}{1 - \delta}$$

in this case. Thus, if $\sigma < 0$ then the denominator changes sign first, while if $\sigma > 0$ then the numerator does. Further, note that the numerator changes sign at $V_{13} = \tilde{V}_{13}$, i.e., at the point where Assumption A4 becomes violated. Finally, note that

$$\lim_{V_{13} \rightarrow \infty} P(1) = -\frac{[1 - \beta(1 - \delta)]\delta}{\beta(1 - \delta)} < 0$$

and thus, at some point as V_{13} increases from zero, we must return to the $0 < \lambda_0 < 1 < \lambda_1$ case (i.e., we must again be to the right of line A).

With the above in mind, suppose $\sigma < 0$. Then $\tau'(V_{13}) > 0$ except at $\bar{V}_{13}$, and further the denominator of ϕ_1 changes sign first. Thus, initially as V_{13} increases from zero, τ moves

rightward in Figure 8 along the line $\Delta = 1/\beta$, keeping it in the region to the right of line A. As V_{13} increases through V_{13} , however, the denominator of ϕ_1 switches signs, while the numerator does not, and thus τ jumps discontinuously to an arbitrarily large negative number. Thus, we jump to the far left in (τ, Δ) -space, to the left of line B (but still above the horizontal axis); that is, we jump into a region featuring $\lambda_0 < -1 < \lambda_1 < 0$. From there, τ continues to increase as V_{13} increases, and as noted above eventually we must end up back to the right line A. Since τ is continuous in V_{13} on $V_{13} > \bar{V}_{13}$, as V_{13} increases from $\bar{V}_{13}$, line B is crossed and the economy move into the $\lambda_0 < \lambda_1 < -1$ region

Suppose instead that $\sigma > 0$. Then $\tau'(V_{13}) < 0$ except at $\bar{V}_{13}$. Further, the numerator of ϕ_1 changes sign before the denominator. This implies that as V_{13} approaches $\bar{V}_{13}$ from below, we must have $\phi_1 \rightarrow -\infty$, and thus $\tau \rightarrow -\infty$. As V_{13} passes through $\bar{V}_{13}$, τ jumps to $+\infty$, putting it back in the $0 < \lambda_0 < 1 < \lambda_1$ region. Since $\tau(\cdot)$ is strictly decreasing and continuous on $V_{13} > \bar{V}_{13}$, if it were to again cross the upward-sloping line it could never return to this region. Since, as argued above, it must be in this region for V_{13} arbitrarily large, this implies that it will not cross the upward-sloping line a second time, i.e., after the jump we remain in the $0 < \lambda_0 < 1 < \lambda_1$ region indefinitely. Putting this together, it implies that, as V_{13} increases from 0, line A is crossed and the economy moves into the $1 < \lambda_0 < \lambda_1$ region. region);

Proof of Proposition 4

Note that

$$\phi_0 = \frac{\sigma}{\beta(\chi - V_{23})}$$

$$\phi_1 = \frac{\theta}{\beta(\chi - V_{23})}$$

Thus, the trace τ and determinant Δ of the evolution matrix are, as functions of V_{23} , given respectively by

$$\tau(V_{23}) = 1 - \delta + \frac{\theta}{\beta(\chi - V_{23})}$$

$$\Delta(V_{23}) = \frac{\chi}{\beta(\chi - V_{23})} = \psi(\tau) \equiv \frac{\chi}{\theta}[\tau - (1 - \delta)]$$

Thus, the possible combinations of τ and Δ must all lie on the line $\Delta = \psi(\tau)$ in (τ, Δ) -space, which has slope $\psi' = \chi/\theta > 0$ and intercept $\psi(0) = -(1 - \delta)\chi/\theta < 0$. We also see that any real value of τ is possible (with the appropriate choice of V_{23}), and further τ is continuous and strictly decreasing in V_{23} everywhere except at the point $V_{23} = \chi < 0$, with τ approaching $+\infty$ as V_{23} approaches χ from above, and τ approaching $-\infty$ as V_{23} approaches χ from below. Given that Δ is a strictly increasing linear function of τ , all of these properties are also inherited by Δ . Note also from (9) that $\psi' < 1$, i.e., the slope of that linear function is less than one. Further,

$$\lim_{V_{23} \rightarrow \infty} \tau(V_{23}) = \lim_{V_{23} \rightarrow -\infty} \tau(V_{23}) = 1 - \delta,$$

$$\lim_{V_{23} \rightarrow \infty} \Delta(V_{23}) = \lim_{V_{23} \rightarrow -\infty} \Delta(V_{23}) = 0.$$

Also, since in these limiting cases the system is to the left of line A in Figure 8, and since $\psi' < 1$, it follows that $\psi(\tau)$ crosses line A—which has slope exactly one—at a point northeast of $(\tau, \Delta) = (1 - \delta, 0)$. We may then define V_{23}^* to be the value of V_{23} at which $\Delta = \psi(\tau)$ intersects line A, V_{23}^{***} the value at which it intersects line B, V_{23}^{**} the value (if it exists) at which it intersects line D, and V_{23}' and V_{23}'' , $V_{23}' < V_{23}''$, the values (if they exist) at which it intersects parabola C.

With these properties in mind, first consider what happens as V_{23} increases beginning from zero. Initially, as found in Proposition 2, we must be in the region to the right of line A and above the horizontal axis in Figure 8, i.e., we are in the $0 < \lambda_0 < 1 < \lambda_1$ region. Since $\tau(\cdot)$ is strictly decreasing and continuous in V_{23} on $V_{23} \geq 0$, and since for V_{23} sufficiently large we must be to the left of line A as argued above, it follows that the system crosses line A at some point as V_{23} increases from zero; that is, we have $V_{23}^* > 0$. Further, since the first eigenvalue transition as we move southwest along the line $\Delta = \psi(\tau)$ can only happen by either crossing line A or crossing the horizontal axis, and since the crossing of line A happens at a point above the horizontal axis (as noted above), the crossing of line A is in fact the first eigenvalue transition. This crossing occurs when the trace is $\tau = \tau^*$, where τ^* is the (unique) solution to $\psi(\tau^*) = \tau^* - 1$. The corresponding value of Δ is then $\Delta^* = \tau^* - 1$. One can verify that

$$\Delta^* = -\frac{\delta\chi}{\chi - \theta}$$

Thus, $\Delta^* < 1$ if and only if $\hat{\sigma} < 0$.

Thus, suppose first that $\hat{\sigma} < 0$. Then the first eigenvalue transition occurs as the line A is crossed, and this happens at a point below line D in Figure 8. As such, the system crosses into the $0 < \lambda_0 < \lambda_1 < 1$ region at this point, i.e., λ_1 must decrease through one. This confirms the first part of the proposition.

Suppose instead that $\hat{\sigma} > 0$. Then the crossing of line A (i.e., the first transition) occurs at a point above line D in the Figure. As such, the system crosses into the $1 < \lambda_0 < \lambda_1$ region, i.e., λ_0 must increase through one. This confirms the second part of the proposition.

Finally, consider now what happens as V_{23} decreases from zero, which (except for at the discontinuity point $V_{23} = \chi$) causes the system to move northeast along the line $\Delta = \psi(\tau)$. Since we are beginning from a point to the right of line A and above the horizontal axis, and since as already established the intersection with line A happens at a value $V_{23} > 0$, as long as $V_{23} > \chi$ the system remains in that region as V_{23} decreases from zero, i.e., we continue to have $0 < \lambda_0 < 1 < \lambda_1$. At the point $V_{23} = \chi$, the system loses a dimension, and from (11) we see that the lone eigenvalue is χ/θ , which as noted above is between 0 and 1. For V_{23} just below χ , τ jumps to $-\infty$, which, since the lone intersection of $\Delta = \psi(\tau)$ with line A occurs at a point with $\tau > 0$, places the system in the far southwest corner of Figure 8, but above line A, so that we are in the region with $\lambda_0 < -1$, $0 < \lambda_1 < 1$. As V_{23} continues to decrease, so that we continue to move northeast along $\Delta = \psi(\tau)$, the next transition to occur happens either when line A, line B, or the horizontal axis is crossed. As just noted, the intersection with line A occurs for $\tau > 0$, and thus from Figure 8 we see clearly that line B would already have been crossed at that point, and thus a crossing of line A cannot be the next transition. Further, since the system approaches $(\tau, \Delta) = (1 - \delta, 0)$ from the southwest as $V_{23} \rightarrow -\infty$, and since τ is continuous in V_{23} on $V_{23} < \chi$, the horizontal axis will never be crossed. Thus, the next transition must be the crossing of line B, which happens

at $V_{23} = V_{23}^{***}$ and puts the system into the region with $-1 < \lambda_0 < 0 < \lambda_1 < 1$. The system then remains in this region as V_{23} decreases without bound. This confirms the third part of the proposition.

Figure 8: Eigenvalue configurations: trace-determinant space

